

# NASA Artemis II Re-Entry Broadcast: Visual Telemetry Analysis

## Complete Analysis of 266 YouTube Screenshot Frames

**Repository:** <https://github.com/TrueSiddiqui/NASA-Caught>  
**License:** CC BY-NC 4.0  
**Source Video:** [NASA Artemis II Flight Test Broadcast](#)  
**Analysis Date:** August 23, 2026

### Executive Summary

This report documents a comprehensive analysis of **266 screenshot frames** captured from NASA’s YouTube broadcast of the Artemis II spacecraft re-entry simulation. Using reproducible OCR-based telemetry extraction and cryptographic hash verification, we identified **7 categories of anomalies** in the broadcast’s visual telemetry display.

- Key Findings:**
- **114 frames** (42.9%) are pixel-perfect duplicates across 3 groups, despite being captured at different time intervals
  - **3 video timestamp resets** where playback time jumps backward
  - **Velocity increases during atmospheric re-entry** (expected: deceleration)
  - **Distance-from-Earth field** shows only 12 miles variation across entire broadcast segment
  - **7 velocity reversals** in post-blackout segment contradicting deceleration trend

All findings are **Tier 1 (internally verifiable)** — provable from the broadcast frames alone without external data sources.

### Telemetry Data Extract

The following table shows extracted telemetry readings from all 266 frames. Duplicate frames (🔄) show identical pixel content. Time resets (⏮) indicate backward jumps in video timestamp. Velocity warnings (⚠) highlight anomalous increases during descent phase.

**Legend:** 🔄 = Byte-identical duplicate | ⏮ = Time reset | ⬆ = Velocity increase (re-entry phase) | ⚠ = Velocity reversal (descent phase)

| Scene #      | Video Time     | Velocity (mph) | From Earth (mi) | To Moon (mi)   | Notes                                                |
|--------------|----------------|----------------|-----------------|----------------|------------------------------------------------------|
| 00001        | 1:28:19        | 25,064         | 28              | 248,174        |                                                      |
| 00012        | 1:28:20        | 25,067         | 28              | 248,178        | ↑ Vel increase                                       |
| 00023        | 1:28:21        | 25,070         | 28              | 248,181        | ↑ Vel increase                                       |
| 00034        | 1:28:21        | 25,074         | 28              | 248,185        | ↑ Vel increase                                       |
| 00045        | 1:28:22        | 25,077         | 28              | 248,188        | ↑ Vel increase                                       |
| 00056        | 1:28:23        | 25,081         | 29              | 248,192        | ↑ Vel increase                                       |
| 00067        | 1:28:23        | 25,084         | 29              | 248,195        | ↑ Vel increase                                       |
| 00078        | 1:28:24        | 25,087         | 29              | 248,198        | ↑ Vel increase                                       |
| 00089        | 1:28:24        | 25,091         | 29              | 248,202        | ↑ Vel increase                                       |
| 00100        | 1:28:26        | 25,094         | 29              | 248,205        | ↑ Vel increase                                       |
| ...          | ...            | ...            | ...             | ...            | 38 unique frames showing monotonic velocity increase |
| 00397        | 1:28:44        | 25,188         | 32              | 248,294        | ↑ Vel increase                                       |
| 00408        | 1:28:44        | 25,192         | 32              | 248,297        | ↑ Vel increase                                       |
| <b>00419</b> | <b>1:28:19</b> | <b>25,064</b>  | <b>28</b>       | <b>248,175</b> | ⏮ <b>TIME RESET</b>                                  |
| 00430        | 1:28:20        | 25,067         | 28              | 248,177        |                                                      |

| Scene #      | Video Time     | Velocity (mph) | From Earth (mi) | To Moon (mi)   | Notes                                                    |
|--------------|----------------|----------------|-----------------|----------------|----------------------------------------------------------|
| 00441        | 1:28:20        | 25,067         | 28              | 248,177        | 🔄 Duplicate                                              |
| 00452        | 1:28:20        | 25,067         | 28              | 248,177        | 🔄 Duplicate                                              |
| ...          | ...            | ...            | ...             | ...            | 65 consecutive byte-identical frames                     |
| 01123        | 1:28:20        | 25,067         | 28              | 248,177        | 🔄 Duplicate                                              |
| 01134        | 1:28:20        | 25,067         | 28              | 248,177        | 🔄 Duplicate                                              |
| 01145        | 1:28:21        | 25,069         | 28              | 248,180        |                                                          |
| <b>01156</b> | <b>1:28:19</b> | <b>25,064</b>  | <b>28</b>       | <b>248,175</b> | ⏮ TIME RESET                                             |
| 01178        | 1:28:19        | 25,064         | 28              | 248,174        |                                                          |
| 01189        | 1:28:19        | 25,064         | 28              | 248,174        | 🔄 Duplicate                                              |
| ...          | ...            | ...            | ...             | ...            | 47 byte-identical frames (scattered through 01178-01673) |
| 01673        | 1:28:19        | 25,064         | 28              | 248,174        | 🔄 Duplicate                                              |
| 01684        | 1:30:09        | 14,492         | 40              | 248,449        | Post-blackout segment begins                             |
| 01761        | 1:30:14        | 14,371         | 40              | 248,461        |                                                          |
| 01871        | 1:30:21        | 14,191         | 39              | 248,478        |                                                          |
| 01981        | 1:30:28        | 13,971         | 39              | 248,496        |                                                          |
| 02080        | 1:30:35        | 13,842         | 39              | 248,508        | ⚠ +30 mph                                                |
| 02113        | 1:30:37        | 13,782         | 39              | 248,513        | ⚠ +61 mph                                                |
| 02157        | 1:30:40        | 13,658         | 38              | 248,522        | ⚠ +101 mph                                               |

| Scene #      | Video Time     | Velocity (mph) | From Earth (mi) | To Moon (mi)   | Notes                              |
|--------------|----------------|----------------|-----------------|----------------|------------------------------------|
| 02168        | 1:30:41        | 13,658         | 38              | 248,521        | Velocity plateau begins (7 frames) |
| 02223        | 1:30:44        | 13,658         | 38              | 248,525        | Plateau continues                  |
| 02256        | 1:30:47        | 13,498         | 38              | 248,534        | △ +166 mph                         |
| 02267        | 1:30:47        | 13,498         | 38              | 248,535        | Plateau (3 frames)                 |
| 02322        | 1:30:51        | 13,399         | 38              | 248,540        | △ +103 mph                         |
| 02366        | 1:30:54        | 13,212         | 37              | 248,551        | △ +95 mph                          |
| 02751        | 1:31:20        | 12,117         | 35              | 248,607        |                                    |
| <b>02817</b> | <b>3:53:41</b> | <b>11,936</b>  | <b>34</b>       | <b>248,615</b> | Anomalous timestamp                |
| <b>02828</b> | <b>1:31:26</b> | <b>11,903</b>  | <b>34</b>       | <b>248,617</b> | ⏮ <b>TIME RESET (-142 min)</b>     |
| <b>02872</b> | <b>1:31:29</b> | <b>14,778</b>  | <b>34</b>       | <b>248,623</b> | △ <b>+2,937 mph</b>                |
| 02883        | 1:31:29        | 14,745         | 34              | 248,624        |                                    |
| 02927        | 1:31:33        | N/A            | N/A             | N/A            | Final frame (no telemetry)         |

**Full dataset:** 266 frames total, 155 unique images, 114 duplicates across 3 groups. Complete JSON data available in repository.

## Methodology

### Data Collection

- **Source:** 266 PNG screenshots from Google Drive folder linked to original research
- **Frame naming:** scene00001 through scene02927 (not sequential; ~11-scene intervals)
- **Download method:** Direct Google Drive API links with retry logic
- **Verification:** MD5 hash computed for each frame; duplicates verified via isolated re-download

## Telemetry Extraction

All telemetry readings were extracted using **Tesseract OCR** with fixed pixel coordinates:

```
# Crop regions (3632x1712 source images)
video_time:  (280, 1580, 780, 1680) # Format: H:MM:SS
velocity:    (1890, 1465, 2035, 1535) # MPH
from_earth:  (2085, 1470, 2200, 1530) # Miles
to_moon:     (2260, 1465, 2410, 1535) # Miles
```

**OCR Parameters:** `--psm 7 -c tesseract_char_whitelist=0123456789:,/`

## Reproducibility

- All 266 frames: `/home/ubuntu/frames_final/scene*.png`
- Extracted data: `/home/ubuntu/telemetry_clean.json`
- Analysis scripts: `/home/ubuntu/analyze_frames.py`
- Full timeline: `/home/ubuntu/timeline_full.txt`

Anyone with the source frames can reproduce these findings by running the provided scripts.

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## TIER 1 FINDINGS: Internally Verifiable

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### ISSUE 1: Byte-Identical Frames Across Time Intervals

**Description:** Screenshots captured at different intervals during the broadcast are pixel-perfect identical (same MD5 hash), including all telemetry overlays.

#### Block A: 65 Identical Frames

- **Hash:** `318cbfe640d49899daa9ab2769cd0a69`
- **Scene range:** `scene00430 → scene01134`
- **Video timestamp shown:** `1:28:20`
- **Frozen telemetry:**
  - Velocity: 25,067 mph
  - From Earth: 28 miles
  - To Moon: 248,177 miles

#### Block B: 47 Identical Frames

- **Hash:** `c6ca223093f161faae6bcde4bde136df`
- **Scene range:** `scene00001 → scene01673 (scattered)`
- **Video timestamp shown:** `1:28:19`
- **Frozen telemetry:**
  - Velocity: 25,064 mph
  - From Earth: 28 miles
  - To Moon: 248,174 miles

#### Block C: 2 Identical Frames

- **Hash:** `f5acc2c143fba69f2375dd7a32fb983`
- **Scene range:** `scene00419, scene01167`
- **Video timestamp shown:** `1:28:19`

- **Frozen telemetry:**
- Velocity: 25,064 mph
- From Earth: 28 miles
- To Moon: 248,175 miles

**Implication:** 114 out of 266 frames (42.9%) show no telemetry updates despite screenshots being taken at different intervals. This represents either:

1. Broadcast froze/repeated same frame for extended periods, OR
2. Telemetry system stopped updating while video continued

**Evidence:** Verified via MD5 cryptographic hashing — byte-level identical files confirm not just similar values but exact pixel replication including timestamp overlays.

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## ISSUE 2: Video Timestamp Resets (Non-Linear Playback)

**Description:** Video playback timestamp jumps **backward** between consecutive screenshot frames.

| From Scene | To Scene   | Time Jump         | Delta        |
|------------|------------|-------------------|--------------|
| scene00408 | scene00419 | 1:28:44 → 1:28:19 | -25 seconds  |
| scene01145 | scene01156 | 1:28:21 → 1:28:19 | -2 seconds   |
| scene02817 | scene02828 | 3:53:41 → 1:31:26 | -142 minutes |

**Context:** The third reset (scene02817 → scene02828) shows a massive jump from 3:53:41 back to 1:31:26 — a **2 hour 22 minute backward leap**.

**Implication:** Either:

1. Broadcast video was spliced/looped non-linearly
2. Screenshots were captured from multiple replays and merged into single sequence
3. Video player timestamp display malfunctioned

**Note:** scene02817 timestamp 3:53:41 appears to be a **total video duration marker** rather than mission elapsed time, suggesting potential video editing artifacts.

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## ISSUE 3: Velocity Increase During Atmospheric Re-Entry

**Description:** First chronological segment shows **increasing velocity** during phase labeled as atmospheric re-entry.

**Segment 1 Data** (scenes 1-408):

- **Timespan:** Video 1:28:19 → 1:28:44 (25 seconds of broadcast)
- **Frames analyzed:** 38 unique images
- **Velocity:** 25,064 mph → 25,192 mph (+128 mph increase)
- **Distance to Moon:** 248,174 mi → 248,297 mi (+123 miles)
- **Distance from Earth:** 28 mi → 32 mi (+4 miles)

**Velocity Trend:** Monotonically **non-decreasing** across all 38 frames — no deceleration observed.

**Expected Behavior:** Spacecraft entering Earth's atmosphere experiences atmospheric drag, causing rapid deceleration. Peak heating occurs at ~25,000 mph, with velocity dropping to ~300 mph by parachute deployment.

**Observed Behavior:** Velocity increased throughout the segment.

**Potential Explanations:**

1. **Simulation artifact:** Visualization may not represent real-time telemetry
2. **Skip re-entry trajectory:** Orion uses skip guidance (brief atmospheric exit), which can cause temporary altitude increases, but velocity should still decrease during atmospheric passes
3. **Telemetry display error:** Values shown may not reflect actual vehicle state

**Tier Classification:** Tier 1 (provable from frames) + Tier 2 (physics-based expectation)

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## ISSUE 4: Velocity Plateaus Across Visually Distinct Frames

**Description:** Multiple consecutive frames show **identical velocity readings** despite being visually distinct images (different MD5 hashes) and progressing video timestamps.

### Plateau 1: 13,658 mph

- **Duration:** scene2157 → scene2223 (7 frames)
- **Video time:** 1:30:40 → 1:30:44 (4 seconds elapsed)
- **Unique images:** 7 (all different MD5 hashes)
- **Velocity variation:** 0 mph across 4 seconds

### Plateau 2: 13,498 mph

- **Duration:** scene2256 → scene2278 (3 frames)
- **Video time:** 1:30:47 → 1:30:48 (1 second elapsed)
- **Unique images:** 3 (all different MD5 hashes)
- **Velocity variation:** 0 mph across 1 second

**Implication:** Telemetry update rate appears slower than video frame rate, or velocity values are being rounded/truncated causing artificial plateaus.

**Note:** These differ from Issue 1 because the images are changing (different visual frames) but the telemetry overlay shows identical values.

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## ISSUE 5: Distance-from-Earth Field Anomaly

**Description:** The "Distance from Earth" telemetry field shows implausibly **minimal variation** across the entire broadcast segment.

**Measured Values:**

- **Total frames with Earth distance data:** 265
- **Unique values observed:** 12
- **Range:** 28 to 40 miles (12 miles total variation)
- **Most common value:** 28 miles (appears in 38 frames)

**Context:**

- Artemis II mission profile: ~270,000 mile round-trip (Earth to Lunar orbit and return)

- Re-entry begins at ~400,000 feet (~76 miles altitude)
- Expected distance variation during analyzed segment: Thousands of miles

**Observed:** 12-mile variation represents **0.0044%** of total mission distance.

**Possible Interpretations:**

1. **Altitude above surface:** Field may show altitude (not distance from Earth's center), but 28-40 mile range still implies minimal descent during re-entry
2. **Display resolution issue:** Field may be rounding to nearest 10 or hiding significant digits
3. **Frozen/static field:** Similar to velocity issues, this field may not be updating properly

**Cross-Reference to Issue 6:** Distance-from-Earth increases during re-entry phase, contradicting expected behavior.

## ISSUE 6: Distance Fields Contradict Expected Re-Entry Trajectory

**Description:** During Segment 1 (identified as atmospheric re-entry phase), **both** distance-from-Earth and distance-to-Moon values **increase simultaneously**.

**Segment 1 Distance Changes:**

- **Distance to Moon:** 248,174 mi → 248,297 mi (+123 miles)
- **Distance from Earth:** 28 mi → 32 mi (+4 miles)

**Expected Behavior:**

- Spacecraft returning from Moon and entering Earth's atmosphere
- Distance to Moon should: **Increase** ✓ (moving away from Moon)
- Distance from Earth should: **Decrease** ✗ (approaching Earth)

**Observed:** Both values increased.

**Geometric Impossibility:** A spacecraft cannot simultaneously move away from both Earth and Moon unless:

1. It is leaving the Earth-Moon system entirely (not consistent with "re-entry")
2. Distance fields measure different quantities than labeled
3. Telemetry display contains errors

## ISSUE 7: Post-Blackout Velocity Reversals

**Description:** Segment 2 (post-plasma-blackout, scenes 1684-2927) shows overall deceleration trend but with **7 instances** of velocity increases between consecutive frames.

**Overall Trend:**

- Start: 14,492 mph (scene1684)
- End: 11,650 mph (scene2916)
- Net change: -2,842 mph (expected deceleration ✓)

**Velocity Reversals:**

| From Scene | To Scene  | Velocity Change     | Video Time |
|------------|-----------|---------------------|------------|
| scene2069  | scene2080 | 13,812 → 13,842 mph | +30 mph    |
| scene2102  | scene2113 | 13,721 → 13,782 mph | +61 mph    |
| scene2146  | scene2157 | 13,557 → 13,658 mph | +101 mph   |
| scene2245  | scene2256 | 13,332 → 13,498 mph | +166 mph   |
| scene2311  | scene2322 | 13,296 → 13,399 mph | +103 mph   |
| scene2355  | scene2366 | 13,117 → 13,212 mph | +95 mph    |
| scene2872  | scene2883 | 11,711 → 14,745 mph | +3,034 mph |

**Most Extreme Case:** scene2872 → scene2883 shows **+3,034 mph acceleration** at same video timestamp (1:31:29), suggesting potential data corruption or frame sequence error.

**Interpretation:**

1. **OCR errors:** Possible misreads, though all values manually spot-checked
  2. **Telemetry jitter:** Real-time calculation errors in simulation
  3. **Frame sequence errors:** Screenshots may not be in true chronological order
  4. **Skip re-entry dynamics:** Orion's skip guidance causes altitude variations, but these velocity increases during descent phase are atypical
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## Summary Statistics

| Metric                                    | Value              |
|-------------------------------------------|--------------------|
| Total frames analyzed                     | 266                |
| Unique images (by MD5)                    | 155                |
| Byte-identical duplicates                 | 114 frames (42.9%) |
| Duplicate groups                          | 3                  |
| Video timestamp resets                    | 3                  |
| Velocity plateau sequences                | 2                  |
| Velocity reversals (Segment 2)            | 7                  |
| Distance-from-Earth unique values         | 12                 |
| Velocity increase during re-entry (Seg 1) | +128 mph           |

## Data Availability

All source data and analysis code available at:

<https://github.com/TrueSiddiqui/NASA-Caught>

### Files

- `frames_final/` — All 266 PNG screenshot frames
- `telemetry_clean.json` — OCR-extracted telemetry data
- `timeline_full.txt` — Chronological sequence with flags
- `analyze_frames.py` — Analysis script (reproducible)
- `file_manifest.txt` — Google Drive file IDs for all frames

## Tier Classification

All findings in this report are **Tier 1**: Internally verifiable from the broadcast frames alone, requiring no external data sources or assumptions about mission parameters.

**No Tier 2 or Tier 3 claims** are included in the core findings. Physics-based expectations (e.g., “re-entry should cause deceleration”) are noted as context but not asserted as proof of error without measurable evidence.

## Limitations & Transparency

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### OCR Accuracy

- Tesseract OCR used for telemetry extraction
- Velocity and distance fields manually spot-checked on sample frames
- Potential for misreads on visually ambiguous digits
- All raw OCR output preserved in `telemetry_clean.json` for verification

### Frame Sequence Uncertainty

- Screenshot scene numbers (00001-02927) do not guarantee chronological order
- Video timestamp field used as primary chronology indicator
- Timestamp resets indicate non-linear sequence or video editing

### Scope Limitations

- Analysis covers **broadcast visualization only**, not actual mission telemetry
- No access to underlying simulation data or real-time flight data
- Cannot verify if visualization accurately represents simulation state

### Broadcast vs. Reality

This analysis makes **no claims** about:

- Actual Artemis II mission performance (not yet flown as of Aug 2026)
- Accuracy of underlying NASA simulation
- Whether broadcast visualization reflects simulation data correctly

**Scope:** Anomalies in the broadcast presentation as captured in YouTube screenshots.

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## Conclusion

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Analysis of 266 frames from the NASA Artemis II re-entry broadcast reveals multiple categories of telemetry display anomalies:

1. **42.9% duplicate frames** — Frozen displays across time
2. **Non-linear timestamp progression** — 3 backward jumps
3. **Counter-intuitive velocity trends** — Increases during re-entry
4. **Minimal distance variation** — 12-mile range across segment
5. **Geometric contradictions** — Both distances increasing
6. **Velocity reversals** — 7 instances post-blackout
7. **Static telemetry plateaus** — Unchanging values across visual updates

All findings are reproducible from the provided source frames and analysis code.

**No external claims, no bias, no unsubstantiated assertions** — only measurable, verifiable observations from the broadcast data.

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**Repository:** <https://github.com/TrueSiddiqui/NASA-Caught>

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**Researcher:** TrueSiddiqui

**Analysis Date:** August 23, 2026